

# Graphs (With Explanation) - 7

## Figures Generation

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Using:

- Matplotlib
- Seaborn

```
In [1]: import matplotlib.pyplot as plt    # Used for plotting graphs
import seaborn as sns                    # Used for advanced
visualization
import numpy as np                      # Used for generating data
```

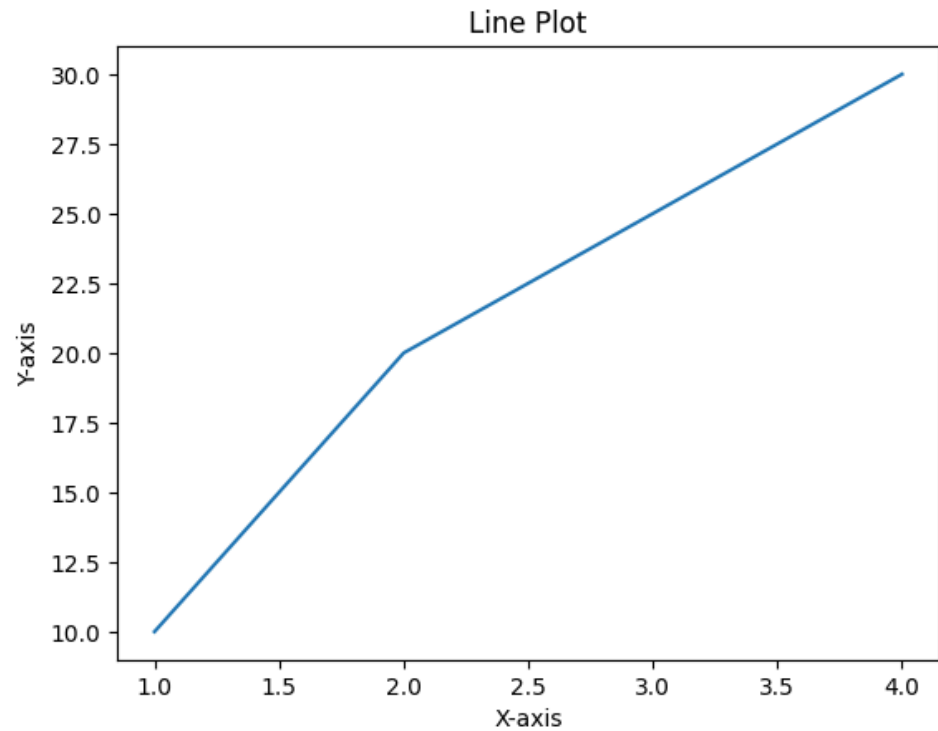
## 1. Line Plot

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Displays data points connected by lines to show trends.

```
In [2]: x = [1, 2, 3, 4]
y = [10, 20, 25, 30]

plt.plot(x, y)
plt.title("Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```

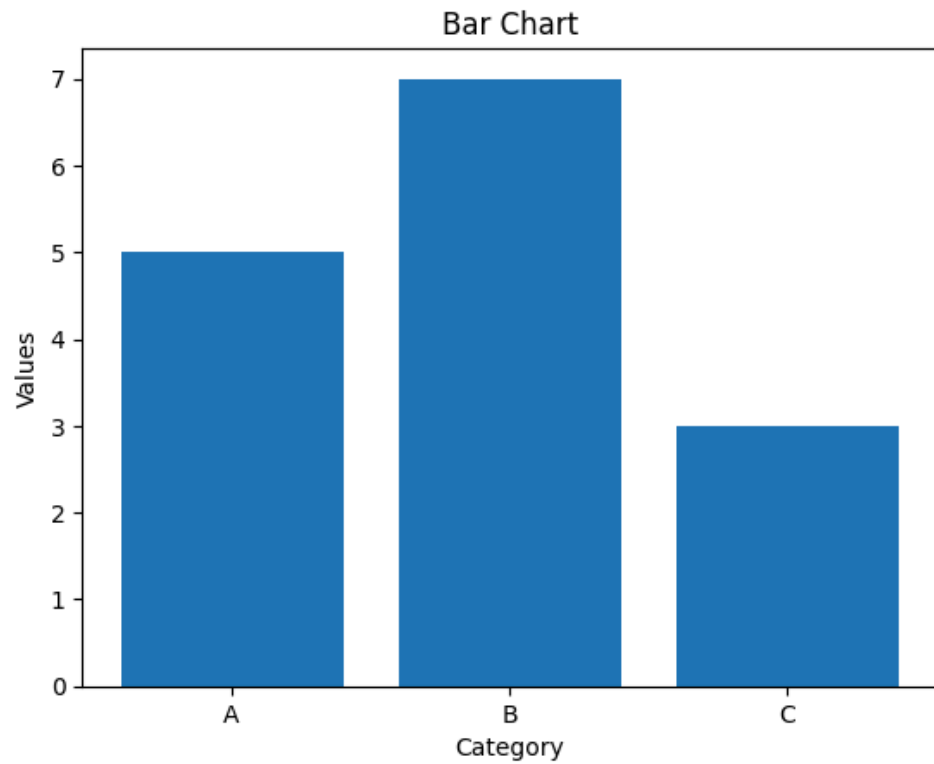


## 2. Bar Chart

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Compares values among different categories.

```
In [3]: categories = ['A', 'B', 'C']  
values = [5, 7, 3]  
  
plt.bar(categories, values)  
plt.title("Bar Chart")  
plt.xlabel("Category")  
plt.ylabel("Values")  
plt.show()
```



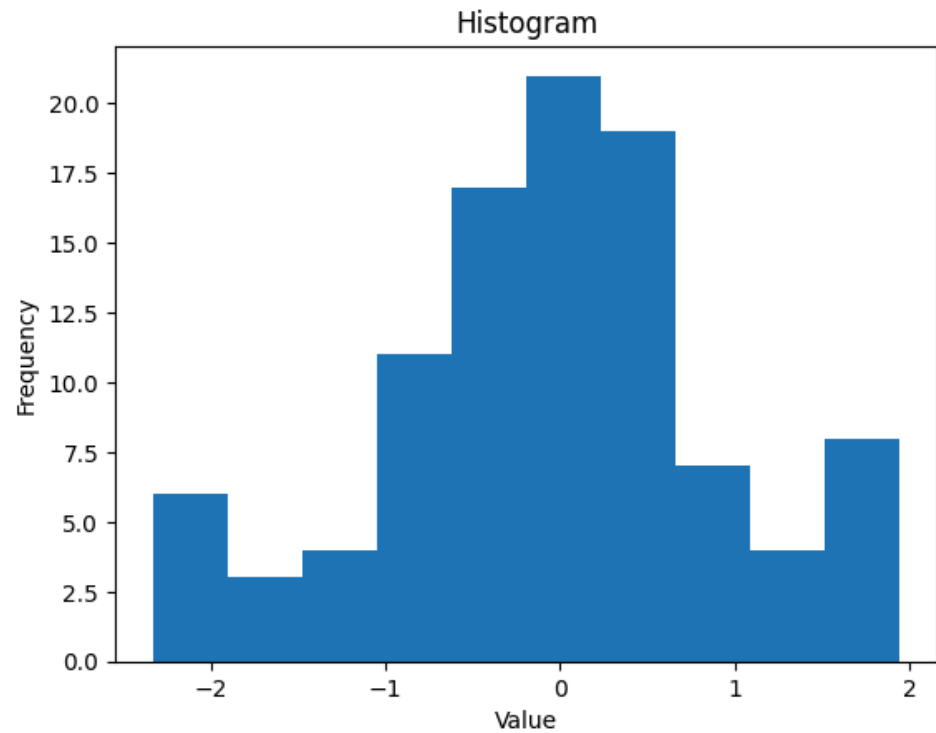
### 3. Histogram

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Shows the frequency distribution of data.

```
In [4]: data = np.random.randn(100)

plt.hist(data)
plt.title("Histogram")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```



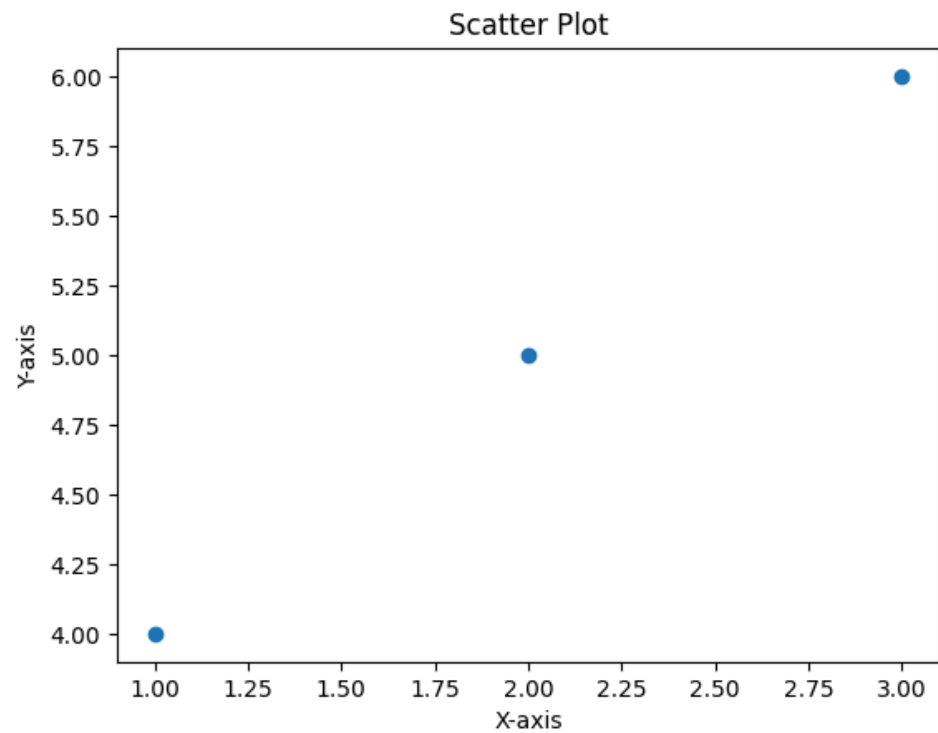
## 4. Scatter Plot

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Shows the relationship between two variables.

```
In [5]: x = [1, 2, 3]
y = [4, 5, 6]

plt.scatter(x, y)
plt.title("Scatter Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```



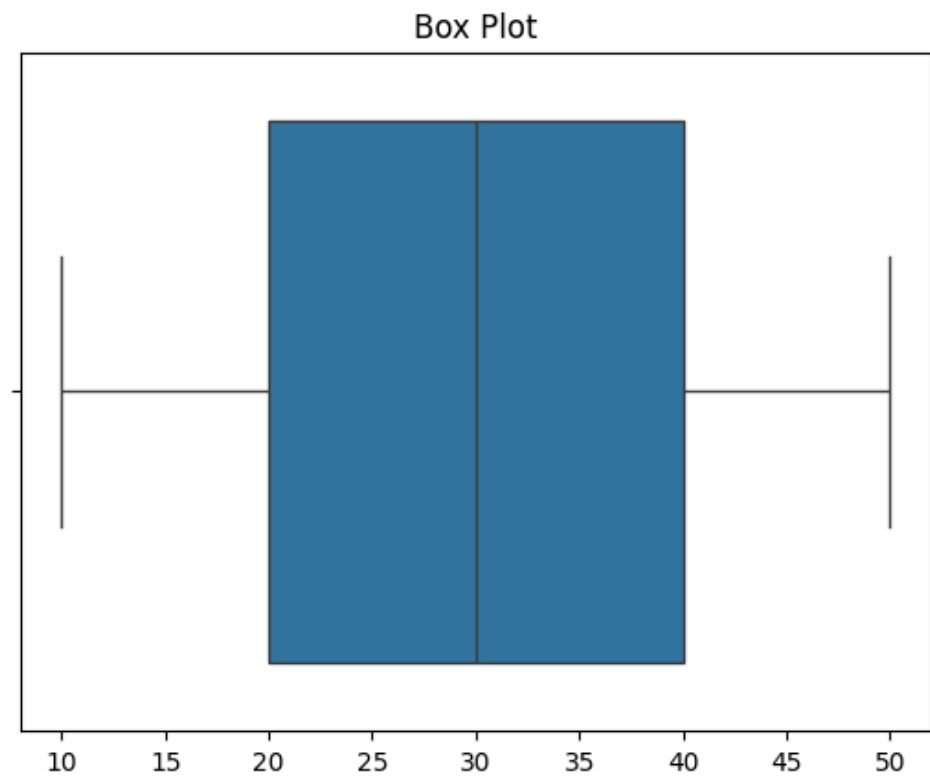
## 5. Box Plot

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Displays data distribution and outliers.

```
In [6]: data = [10, 20, 30, 40, 50]

sns.boxplot(x=data)
plt.title("Box Plot")
plt.show()
```



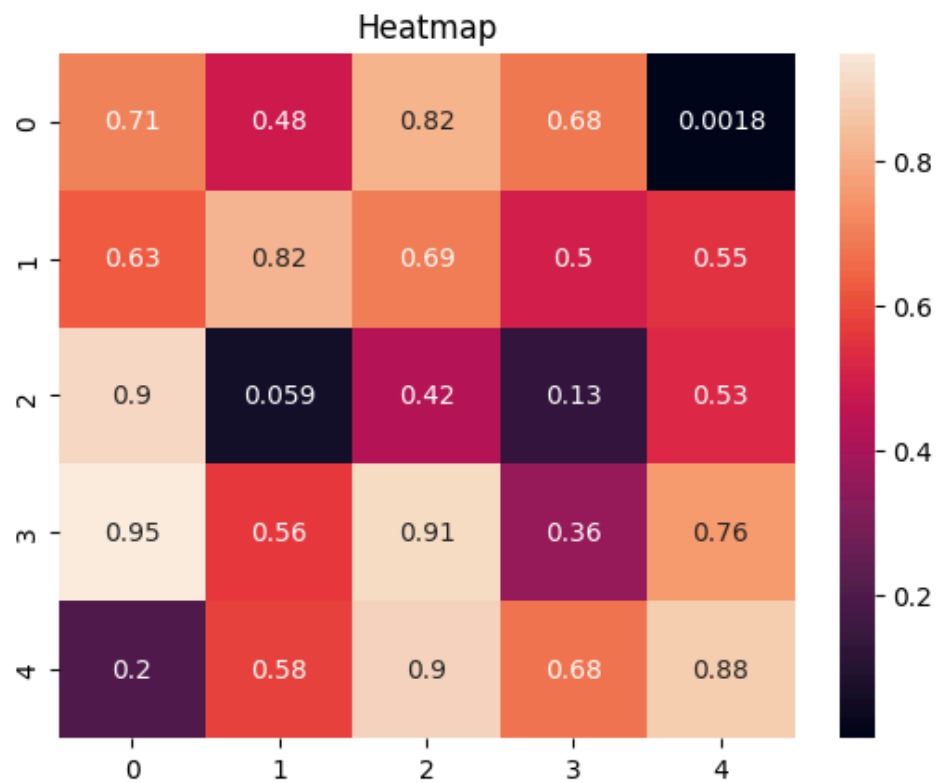
## 6. Heatmap

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Uses colors to represent values in a matrix.

```
In [7]: data = np.random.rand(5,5)

sns.heatmap(data, annot=True)
plt.title("Heatmap")
plt.show()
```



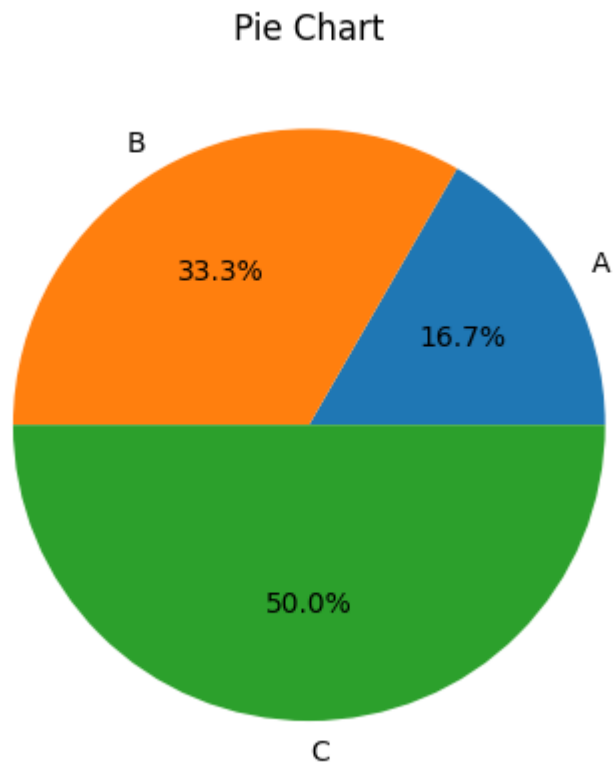
## 7. Pie Chart

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Represents data as portions of a circle.

```
In [8]: values = [10, 20, 30]
labels = ['A', 'B', 'C']

plt.pie(values, labels=labels, autopct='%1.1f%%')
plt.title("Pie Chart")
plt.show()
```



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## Student Information

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**Lab Report No.:** 03